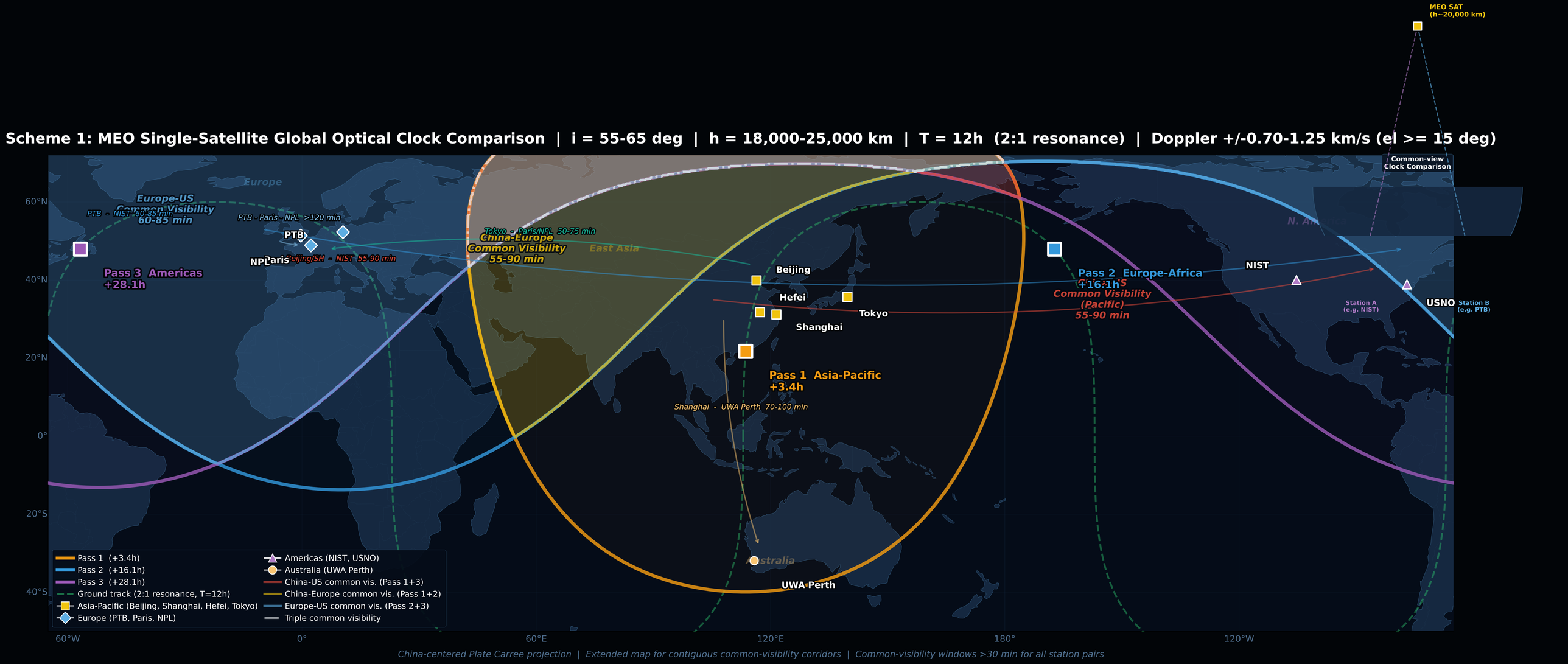
Scheme 1: MEO Single-Satellite Global Optical Clock Comparison | i = 55-65 deg | h = 18,000-25,000 km | T = 12h (2:1 resonance) | Doppler +/-0.70-1.25 km/s (el >= 15 deg)
MEO SAT
(h~20,000 km)
Common-view
Clock Comparison
Europe
Europe-US
Common Visibility
60-85 min
PTB - NIST 60-85 min
PTB - Paris - NPL >120 min
PTB
NPL
Paris
Beijing/SH - NIST 55-90 min
Pass 3 Americas
+28.1h
Tokyo - Paris/NPL 50-75 min
China-Europe
Common Visibility
55-90 min
East Asia
Beijing
Hefei
Tokyo
Shanghai
Pass 1 Asia-Pacific
+3.4h
Shanghai - UWA Perth 70-100 min
Australia
UWA Perth
Pass 2 Europe-Africa
+16.1h
Common Visibility
(Pacific)
55-90 min
NIST
USNO
Station A
(e.g. NIST)
Station B
(e.g. PTB)
60°N
40°N
20°N
0°
20°S
40°S
60°W
0°
60°E
120°E
180°
120°W
Pass 1 (+3.4h)
Pass 2 (+16.1h)
Pass 3 (+28.1h)
Ground track (2:1 resonance, T=12h)
Asia-Pacific (Beijing, Shanghai, Hefei, Tokyo)
Europe (PTB, Paris, NPL)
Americas (NIST, USNO)
Australia (UWA Perth)
China-US common vis. (Pass 1+3)
China-Europe common vis. (Pass 1+2)
Europe-US common vis. (Pass 2+3)
Triple common visibility
China-centered Plate Carree projection | Extended map for contiguous common-visibility corridors | Common-visibility windows >30 min for all station pairs